Functional Annotation Report: Rat *Lalba* (Alpha-lactalbumin, UniProt P00714)

Gene: *Lalba* · **Protein:** Alpha-lactalbumin / Lactose synthase B protein · **Organism:** *Rattus norvegicus* (rat) · **Family:** Glycosyl hydrolase 22 (lysozyme-like fold; IPR001916, IPR000545, IPR023346; Pfam PF00062)

Summary

The rat gene **Lalba** encodes **alpha-lactalbumin**, a small (~14 kDa mature), secreted, calcium-binding milk protein that serves as the **non-catalytic regulatory subunit of lactose synthase** (EC 2.4.1.22). Its identity is unambiguous and matches the UniProt record precisely: the protein belongs to the glycosyl hydrolase family 22 (GH22) / lysozyme-like superfamily, carrying the Lactalbumin (IPR000545) and Glyco_hydro_22 (IPR001916) domains. Despite its lysozyme-like fold, alpha-lactalbumin has no bacteriolytic (glycoside hydrolase) activity of its own. Instead, its primary biochemical function is to **bind the catalytic enzyme β-1,4-galactosyltransferase-1 (B4GALT1 / Gal-T1) in a 1:1 complex and reprogram that enzyme's acceptor-sugar specificity**, switching it from N-acetylglucosamine (GlcNAc) to glucose. The resulting lactose synthase complex transfers galactose from UDP-galactose onto free glucose, producing **lactose (Gal-β1,4-Glc)** + **UDP**, the signature carbohydrate of milk.

Mechanistically, alpha-lactalbumin is a **specifier/modifier protein**: it does not catalyze the glycosyl transfer but lowers the Michaelis constant (Km) of Gal-T1 for glucose by orders of magnitude, binding the "closed" conformation of the enzyme and hydrogen-bonding to glucose to hold it in the acceptor site. The reaction is Mn^{2+}-dependent and proceeds by a random-equilibrium kinetic mechanism. The complex assembles and functions on the **luminal face of the trans-Golgi** of lactating mammary alveolar (secretory) epithelial cells, where B4GALT1 is anchored as a type II membrane protein; alpha-lactalbumin, bearing a cleavable signal peptide, is a soluble protein that is subsequently **secreted into milk as a major whey protein**. A high-affinity structural Ca^{2+} ion stabilizes its native two-lobe (α-helical + β-sheet) fold and is required for correct oxidative folding; the apo form is the archetypal "molten globule."

Lalba expression is **mammary-gland specific** and tightly controlled by the lactogenic hormone trio **insulin, cortisol (glucocorticoid), and prolactin**, with prolactin acting through the **prolactin receptor → JAK2 → STAT5** transcriptional axis. Because lactose is the principal osmotic constituent of milk, the amount of active lactose synthase — and therefore the availability of alpha-lactalbumin — is the **rate-defining control point for lactose biosynthesis and hence milk volume**. This report synthesizes eight confirmed findings drawn from 31 reviewed papers into a coherent mechanistic model of Lalba function, localization, and regulation.

Key Findings

Finding 1 — Lalba is the regulatory subunit of lactose synthase (EC 2.4.1.22)

Rat alpha-lactalbumin is the **regulatory ("B") component of the two-protein lactose synthase enzyme**. Lactose synthase is a 1:1 complex of a catalytic component, β-1,4-galactosyltransferase-1 (B4GALT1 / β4Gal-T1 / Gal-T1), and the regulatory component, alpha-lactalbumin (LA). On its own, B4GALT1 is a galactosyltransferase that transfers galactose from UDP-galactose to N-acetylglucosamine (GlcNAc) residues on glycoproteins and glycolipids. When alpha-lactalbumin binds, it **promotes the binding of glucose to B4GALT1 and alters the enzyme's sugar-acceptor specificity from GlcNAc to glucose**, enabling the complex to synthesize lactose, the major carbohydrate of milk.

Crucially, **alpha-lactalbumin is itself catalytically inactive** — it is a specifier/modifier, not an enzyme. High-resolution (2 Å) crystal structures of the lactose synthase complex (Ramakrishnan & Qasba, 2001) show that LA binds the **closed conformation** of Gal-T1 and holds glucose by hydrogen bonding to its O-1 hydroxyl, which is what re-tunes the enzyme's acceptor preference and **lowers the Km of Gal-T1 for glucose**. The net reaction catalyzed by the complex is:

> **UDP-galactose + D-glucose → lactose (Gal-β1,4-Glc) + UDP**

This directly answers "what reaction is catalyzed, and what is the substrate specificity": the catalytic machinery is B4GALT1, the donor substrate is UDP-galactose, the acceptor substrate (made accessible by alpha-lactalbumin) is free glucose, and the product is lactose. Alpha-lactalbumin's contribution is to define the acceptor specificity.

> *Evidence* — *PMID: 11419947:* "The lactose synthase (LS) enzyme is a 1:1 complex of a catalytic component, beta1,4-galactosyltransferse (beta4Gal-T1) and a regulatory component, alpha-lactalbumin (LA), a mammary gland-specific protein. LA promotes the binding of glucose (Glc) to beta4Gal-T1, thereby altering its sugar acceptor specificity from N-acetylglucosamine (GlcNAc) to glucose, which enables LS to synthesize lactose, the major carbohydrate component of milk."
>
> *Evidence* — *PMID: 9538226:* "alphaLA-Lactalbumin (alphaLA), a major milk protein, is the regulatory subunit of lactose synthase."

Finding 2 — The site of action is the trans-Golgi lumen; the protein is then secreted into milk

Alpha-lactalbumin carries out its function in the **lumen of the trans-Golgi apparatus** of mammary secretory cells. Its catalytic partner B4GALT1 is a **type II trans-Golgi membrane glycosyltransferase** — a short N-terminal cytoplasmic tail, a single transmembrane span, and a stem plus C-terminal catalytic domain that faces the trans-Golgi lumen (Qasba et al., 2008). Because the alpha-lactalbumin-binding catalytic domain projects into the Golgi lumen, the lactose synthase complex necessarily assembles and operates there.

Alpha-lactalbumin itself is a **soluble, secreted protein**: it is synthesized with a cleavable signal peptide, processed to a mature soluble form (recovered in the soluble cellular fraction and chromatographically identical to native milk alpha-lactalbumin), and, after serving its Golgi function, **secreted as one of the major whey proteins of milk**. This defines both the intracellular site of function (trans-Golgi lumen) and the ultimate extracellular destination (milk/whey).

> *Evidence* — *PMID: 18393823:* "It is a trans-Golgi glycosyltransferase (Glyco-T) with a type II membrane protein topology, a short N-terminal cytoplasmic domain, a membrane-spanning region, as well as a stem and a C-terminal catalytic domain facing the trans-Golgi-lumen."
>
> *Evidence* — *PMID: 9538226:* "The expressed alphaLA (ca. 5 microg/g of fresh leaves) was found in the soluble fraction and eluted from a DEAE-Sepharose column in the same salt concentration range as the milk alphaLA."

Finding 3 — Alpha-lactalbumin is a calcium-binding lysozyme homolog that lost bacteriolytic activity

Alpha-lactalbumin and **c-type lysozyme diverged from a common evolutionary ancestor**, sharing amino-acid sequence homology, conserved disulfide bridges, intron–exon gene organization, and three-dimensional fold (McKenzie, 1996). This explains the UniProt family assignment (GH22 / lysozyme-like superfamily) even though alpha-lactalbumin functions as a **non-catalytic modifier** rather than a hydrolase. Where most c-type lysozymes retain bacteriolytic (glycoside-hydrolase) activity, alpha-lactalbumin has lost it and acquired the specifier role.

A defining biochemical property is that **all alpha-lactalbumins strongly bind Ca(II)**, whereas only some c-type lysozymes do. The strong calcium site is formed by the carboxylate side chains of **Asp82, Asp87, and Asp88** together with the backbone carbonyl oxygens of residues 79 and 84. This site is functionally important: CNBr cleavage that disrupts the calcium-binding loop abolishes strong Ca^{2+} binding and **reduces lactose-synthesis activity ~40-fold**, while a secondary Ca^{2+} site preserves partial activity (Berliner et al., 1991). Alpha-lactalbumin is also a classic experimental model for the partially folded **"molten globule"** state.

> *Evidence — PMID: 8765309:* "It has the unusual property of acting as a modifier protein to modify the action of galactosyl transferase to a lactose synthase. Both proteins have diverged from a common ancestor." and "all alpha-lactalbumins strongly bind Ca(II), but only some c-type lysozymes do so."
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> *Evidence — PMID: 1918521:* "The strong calcium-binding site of alpha-lactalbumin comprises the carboxylate side chains of aspartic acid 82, 87, and 88 and the carbonyl oxygens of residues 79 and 84."

Finding 4 — Expression is mammary-specific and induced by insulin, cortisol, and prolactin

Lalba is expressed in a **mammary-gland-restricted** manner and is induced by the classical **lactogenic hormone trio: insulin, cortisol (glucocorticoid), and prolactin**. In rat mammary explant culture, alpha-lactalbumin mRNA is induced by combinations of these three hormones. Consistent with hormonal control of lactogenesis, late-pregnant rats subjected to uteroplacental insufficiency (and the attendant progesterone withdrawal) show elevated

alpha-lactalbumin mRNA, interpreted as premature lactogenesis (O'Dowd et al., 2008). Across the literature, alpha-lactalbumin is repeatedly described as a "mammary gland-specific protein."

> *Evidence — PMID: 18462607:* "Mammary explants were cultured with combinations of insulin, cortisol and prolactin and analysed for alpha-lactalbumin and beta-casein gene expression."
>
> *Evidence — PMID: 11419947:* "a regulatory component, alpha-lactalbumin (LA), a mammary gland-specific protein."

Finding 5 — Substrates and mechanism: UDP-galactose + glucose → lactose, via a Mn^{2+}-dependent random-equilibrium mechanism

Detailed kinetic analysis of milk galactosyltransferase shows that lactose synthesis proceeds by a **random-equilibrium addition of substrates**: UDP-galactose and the acceptor can bind in either order, and alpha-lactalbumin can bind to either the enzyme-Mn^{2+}-acceptor complex or the enzyme-Mn^{2+}-UDP-galactose complex (Bell, Beyer & Hill, 1976). The reaction has an absolute requirement for the divalent cation **Mn^{2+}** and transfers galactose from UDP-galactose onto free glucose to yield lactose.

Because **lactose is the major osmotic constituent of milk**, its synthesis is the principal determinant of milk volume: water is drawn osmotically into the secretory vesicles in proportion to the lactose made. The free-glucose acceptor must be transported into the Golgi lumen (the lactose-synthesis compartment), where B4GALT1 and alpha-lactalbumin reside (Haney, 2004). This couples the specifier function of alpha-lactalbumin directly to the physiological output of the lactating gland.

> *Evidence — PMID: 818085:* "alpha-lactalbumin may bind to either an enzyme-Mn2+-acceptor complex or an enzyme-Mn2+-UDP-galactose complex, suggesting that lactose synthesis also proceeds by a random equilibrium addition of substrates and alpha-lactalbumin."
>
> *Evidence — PMID: 15384581:* "Lactose is the major carbohydrate of human milk and is also the major osmotic constituent of human milk. Therefore, synthesis of lactose is the major determinant of the volume of milk produced by the lactating human mammary gland. Lactose is synthesized from free glucose and UDP-galactose."

Finding 6 — Identity verification: authentic rat mammary alpha-lactalbumin is a distinct, bona fide gene product

A dedicated verification step confirmed that **genuine rat mammary alpha-lactalbumin cDNA and protein sequences exist** and that the Lalba gene product is not to be confused with other galactosyltransferase-modifying proteins. Specifically, an 18-kDa androgen-regulated galactosyltransferase-modifying protein of the rat caput epididymidis has **no primary structural similarity** to rat milk alpha-lactalbumin, and its modifier activity is only ~1/100th that of authentic milk alpha-lactalbumin (Moore, Hall & Hamilton, 1990). This confirms correct identity attribution: the protein under study is the authentic rat mammary alpha-lactalbumin (P00714), matching the UniProt description, organism (*Rattus norvegicus*), and GH22/lactalbumin domain architecture.

> *Evidence — PMID: 2125511:* "there is no primary structural similarity between it and authentic rat mammary alphalactalbumin."

Finding 7 — Two-lobe, calcium-stabilized fold presents a charged surface that forms the lactose synthase interface

X-ray structures of apo- and holo-alpha-lactalbumin (2.2 Å) reveal a native fold consisting of an **α-helical lobe and a β-sheet lobe separated by a cleft**, with a high-affinity Ca^{2+} site. **Ca^{2+} binding stabilizes the native structure and is required for efficient oxidative folding** to the correctly disulfide-bonded native protein, which explains why the apo form collapses into a molten globule (Chrysina, Brew & Acharya, 2000). A **secondary Ca^{2+} site** lies ~7.9 Å from the primary site (Chandra, Brew & Acharya, 1998).

Critically for function, these cation-binding sites cluster on **one charged face of the molecule**, and the proximity of the manganese- and calcium-binding regions together with the location of the functional site on that charged surface implicates this face as the **interface for forming the lactose synthase complex with B4GALT1**. This provides a structural rationale for how a non-catalytic protein docks onto and modifies the galactosyltransferase.

> *Evidence — PMID: 10896943:* "High affinity binding of Ca(2+) to alpha-lactalbumin (LA) stabilizes the native structure and is required for the efficient generation of native protein with correct disulfide bonds from the reduced denatured state." and "the protein is in a similar global native conformation consisting of alpha-helical and beta-subdomains separated by a cleft."
>
> *Evidence — PMID: 9537992:* "The proximity of the manganese and calcium binding region and the location of the functional site on one side of the charged surface of the alpha-lactalbumin molecule suggest that these binding sites might play a role in the formation of the lactose synthase complex."

Finding 8 — Prolactin → JAK2 → STAT5 signaling controls Lalba transcription during lactation

The hormonal induction of *Lalba* operates through the **prolactin receptor → JAK2 → STAT5** transcriptional axis. Conditional repression of STAT5 in the lactating mouse mammary gland **reduced alpha-lactalbumin expression by ~23%** (and whey acidic protein, WAP, by 39%), with impaired alveolar occupancy and 29–41% reduced neonate weight gain; notably, β-casein was comparatively less affected (Reichenstein, Rauner & Barash, 2011). This places alpha-lactalbumin genetically downstream of prolactin signaling and is fully consistent with its induction by the insulin/cortisol/prolactin trio (Finding 4).

> *Evidence — PMID: 21688337:* "A decrease of 39% and 23% in WAP and α-lactalbumin expression, respectively, with no associated effects on β-casein."

Mechanistic Model and Interpretation

Alpha-lactalbumin sits at the convergence of **structure, enzymology, and endocrine regulation** to perform one precise job: committing the mammary gland to lactose (and therefore milk) production. The following model integrates the eight findings.

The core biochemical switch

```
            Without alpha-lactalbumin              With alpha-lactalbumin (Lactose Synthase)
            -------------------------              -----------------------------------------
Enzyme:     B4GALT1 (Gal-T1) alone                 B4GALT1 • alpha-lactalbumin  (1:1 complex)
Donor:      UDP-galactose                          UDP-galactose
Acceptor:   GlcNAc (on glycoproteins)              free D-glucose   (Km for glucose down down)
Cofactor:   Mn2+                                   Mn2+
Product:    Gal-b1,4-GlcNAc (glycan extension)     LACTOSE (Gal-b1,4-Glc) + UDP
Location:   trans-Golgi lumen                      trans-Golgi lumen
```

B4GALT1 is a promiscuous Golgi galactosyltransferase whose default job is to decorate N-glycans with galactose on GlcNAc termini. Alpha-lactalbumin converts this housekeeping enzyme into a dedicated lactose-maker by **physically binding the closed conformation of Gal-T1 and creating/stabilizing a glucose-binding subsite** — hydrogen bonding to the glucose O-1 hydroxyl — which collapses the Km for glucose and makes the otherwise poor acceptor kinetically favored. The transfer chemistry itself (Mn^{2+}-dependent, random-equilibrium) is unchanged; only the acceptor specificity is redefined.

Spatial logic: everything happens in the trans-Golgi lumen

```
Mammary alveolar (secretory) cell
+-------------------------------------------+
|  Cytoplasm                                |
|     glucose --(transporter)--+            |
|                              v            |
|   === trans-Golgi lumen ================  |
|   |  B4GALT1 (membrane-anchored, type II)| |
|   |     +  alpha-lactalbumin (soluble)   | |
|   |     +  UDP-galactose  +  Mn2+        | |
|   |            |                         | |
|   |            v                         | |
|   |         LACTOSE  --> secretory       | |
|   |         (osmotic)    vesicle --> MILK| |
|   ======================================  |
|   alpha-lactalbumin also secreted --> WHEY |
+-------------------------------------------+
```

B4GALT1's catalytic domain faces the Golgi lumen, so the complex must assemble there. Glucose is imported into the lumen as the acceptor. Lactose, being osmotically active and membrane-impermeant, draws water into secretory vesicles — which is why **lactose synthesis is the rate-limiting determinant of milk volume**. Alpha-lactalbumin, having done its catalytic-partner duty, exits with the secretion as a major whey protein.

Structural enablement

The two-lobe lysozyme-like fold (α-helical + β-sheet lobes around a cleft) is held together by a high-affinity structural Ca^{2+} ion coordinated by Asp82/87/88 and backbone carbonyls. Ca^{2+} is a **folding chaperone**: it stabilizes the native disulfide-bonded state and prevents collapse to the molten globule. The cation-binding face is also the **functional surface** that docks onto Gal-T1 near its Mn^{2+} site. Thus calcium does double duty — structural integrity and positioning of the functional interface.

Regulatory logic: a committed, hormone-gated control point

Because B4GALT1 is constitutively present in many tissues but alpha-lactalbumin is **mammary-specific and hormone-gated**, the availability of alpha-lactalbumin is what *licenses* lactose synthesis. The gene is switched on by the lactogenic trio (insulin + cortisol + prolactin), with prolactin signaling through **PRLR → JAK2 → STAT5** to drive transcription. This makes Lalba the **committed, rate-defining step** of lactose biosynthesis: no alpha-lactalbumin → no lactose → no milk volume.

Layer	Component	Role
Hormonal trigger	Insulin + Cortisol + Prolactin	Induce *Lalba* transcription during lactogenesis
Signal transduction	PRLR → JAK2 → STAT5	Prolactin's transcriptional effector; STAT5 loss ↓ LA by ~23%
Gene product	Alpha-lactalbumin (soluble, Ca^{2+}-bound)	Specifier subunit
Catalytic partner	B4GALT1 (trans-Golgi, type II membrane)	Glycosyltransferase catalytic subunit
Cofactor	Mn^{2+}	Required divalent cation for transfer
Substrates	UDP-galactose (donor) + glucose (acceptor)	Random-equilibrium binding
Product	Lactose + UDP	Milk sugar; osmotic driver of milk volume
Site	trans-Golgi lumen → secretion	Function then secretion into whey

Broader specifier principle

The modifier capability of alpha-lactalbumin is evolutionarily remarkable: related β4-GalNAc-transferases from invertebrates (snail *Lymnaea stagnalis*) and bovine mammary gland can also be induced by alpha-lactalbumin to act on glucose (yielding lactose analogs such as GalNAc-β1,4-Glc), showing that the protein recognizes a conserved structural feature of the β4-GalT family rather than a single enzyme. This underscores that alpha-lactalbumin's primary function is **specificity reprogramming of a glycosyltransferase**, not catalysis per se.

Evidence Base

PMID	Finding(s)	Contribution
11419947	F1, F4	Crystal structure of lactose synthase; defines LA as regulatory subunit that switches acceptor from GlcNAc to glucose; "mammary gland-specific"
9538226	F1, F2	States LA is the regulatory subunit of lactose synthase; shows LA is soluble, signal-peptide-processed, secreted
18393823	F2	Establishes B4GALT1 as type II trans-Golgi enzyme with lumen-facing catalytic domain → defines subcellular site
8765309	F3	Evolutionary divergence from lysozyme; modifier role; universal Ca(II) binding
1918521	F3	Defines calcium-binding loop residues (Asp82/87/88); links Ca binding to activity
18462607	F4	Rat mammary explants: insulin/cortisol/prolactin induce LA gene expression
818085	F5	Kinetic mechanism: Mn^{2+}-dependent random-equilibrium; where LA binds
15384581	F5	Substrates (glucose + UDP-galactose); lactose as osmotic driver of milk volume
2125511	F6	Identity verification: authentic rat mammary LA distinct from epididymal GT-modifier
10896943	F7	Apo/holo structures: two-lobe fold; Ca^{2+} stabilizes native fold/folding
9537992	F7	Secondary Ca^{2+} site; charged cation-binding face as lactose synthase interface
21688337	F8	STAT5 conditional repression ↓ LA ~23% → places LA downstream of prolactin/JAK2/STAT5

Supporting / contextual literature: Structural snapshots of Gal-T1 along its kinetic pathway and its open→closed conformational change (PMID: 11485999, PMID: 16497331) corroborate the mechanism by which LA stabilizes the closed, glucose-binding form. Cross-species modifier

studies (PMID: 8816760, PMID: 9949189) show LA can redirect related β4-GalNAc-transferases to glucose, reinforcing its general role as a glycosyltransferase specifier. Metal-ion/molten-globule studies (PMID: 2691213, PMID: 10329172, PMID: 15218531) support the Ca^{2+}-dependent folding narrative. Hormone-regulation papers (PMID: 9513079, PMID: 19383261) confirm LA co-regulation with other milk-protein genes by lactogenic hormones.

Note on off-target literature: Several retrieved papers (e.g., PMID: 30779922 on MIR31/colitis using alpha-lactalbumin only as a nanoparticle material; PMID: 30177023 on IgA nephropathy; PMID: 22933749 on equol) mention "lactalbumin" incidentally and do **not** bear on Lalba function. The cytotoxic "BAMLET/HAMLET"-type complexes (PMID: 30741951, PMID: 20227419) describe a non-native, oleic-acid/metal-complexed folding variant with membrane-perturbing activity — an interesting moonlighting property but **not the primary physiological function** and outside the scope of the rat mammary Lalba annotation.

Limitations and Knowledge Gaps

1. **Species of structural/kinetic evidence.** Much of the definitive structural and enzymological work (crystal structures of the lactose synthase complex, kinetic mechanism) was performed on **bovine and human** alpha-lactalbumin and galactosyltransferase, not rat. Given the very high sequence/structural conservation of alpha-lactalbumins and the confirmed existence of an authentic rat mammary LA cDNA/sequence (Finding 6), these mechanisms are reliably transferable to rat, but direct rat-specific structural data were not reviewed here.
2. **Quantitative rat-specific kinetics.** Exact Km/kcat values for the *rat* lactose synthase complex, and the precise fold-reduction in glucose Km upon rat LA binding, were not extracted. The qualitative mechanism (Km lowering, Mn^{2+}-dependence, random equilibrium) is established, but rat-specific constants remain a gap.
3. **STAT5 effect size is modest.** The ~23% reduction in LA upon STAT5 repression (Finding 8) was measured in **mouse**, and indicates STAT5 is contributory but not the sole regulator — other factors (glucocorticoid receptor, ELF5, and the insulin pathway) likely co-regulate *Lalba*. The relative contribution of each input in rat is not quantified here.
4. **Glucose transport into the Golgi lumen.** The identity and regulation of the transporter that supplies free glucose (the acceptor) to the trans-Golgi lumen in rat mammary cells was referenced conceptually but not resolved at the molecular level.

5. **Moonlighting functions.** Non-canonical properties (molten-globule/oleic-acid cytotoxic complexes; proposed roles in cell-surface carbohydrate modification and cell–cell interactions) are documented for alpha-lactalbumin generally but are not established as physiological functions of rat Lalba and were deliberately deprioritized per the research brief's focus on precise primary function.

Proposed Follow-up Experiments / Actions

1. **Rat-specific structural confirmation.** Solve or model (AlphaFold + experimental validation) the rat alpha-lactalbumin structure and the rat LA·B4GALT1 complex; verify that the Asp82/87/88 calcium loop and the charged complex-forming face are conserved, and compare the interface to the bovine/human complexes.
2. **Rat lactose synthase kinetics.** Reconstitute recombinant rat alpha-lactalbumin with rat B4GALT1 and measure Km(glucose), Km(UDP-galactose), kcat, and Mn^{2+}-dependence ± LA to quantify the specifier effect directly in the native species.
3. **Calcium-loop mutagenesis in rat LA.** Make Asp82/87/88 substitutions and assay (a) Ca^{2+} affinity, (b) folding/molten-globule propensity, and (c) lactose synthase activity, to test the structure→folding→function chain in the rat protein.
4. **Transcriptional dissection in rat mammary cells.** Use reporter assays and ChIP for STAT5, glucocorticoid receptor, and ELF5 on the rat *Lalba* promoter under insulin/cortisol/prolactin to map the regulatory logic and quantify each input's contribution.
5. **Golgi glucose supply.** Identify the Golgi glucose transporter in rat mammary secretory cells and test whether acceptor supply or alpha-lactalbumin abundance is more limiting for lactose output under physiological conditions.
6. **In vivo rate-limiting test.** Modulate rat *Lalba* expression (knockdown / overexpression) in lactating glands and measure lactose content and milk volume to confirm, in rat, that alpha-lactalbumin availability is the rate-defining determinant of milk-sugar synthesis.

Conclusion

Rat *Lalba* encodes alpha-lactalbumin, the **non-catalytic regulatory subunit of lactose synthase (EC 2.4.1.22)**. It binds β-1,4-galactosyltransferase-1 (B4GALT1) in a 1:1 complex and reprograms the enzyme's acceptor specificity from N-acetylglucosamine to glucose (lowering the Km for glucose), so that the Mn^{2+}-dependent complex transfers galactose from UDP-galactose onto glucose to synthesize **lactose**, the principal milk sugar and osmotic driver of milk volume. It acts in the **trans-Golgi lumen** of lactating mammary alveolar cells and is then **secreted as a major whey protein**. A structural Ca^{2+} ion stabilizes its lysozyme-like two-lobe fold (it is a c-type lysozyme homolog that lost hydrolase activity), and its mammary-restricted expression is driven by the lactogenic hormones insulin, cortisol, and prolactin via the **prolactin → JAK2 → STAT5** axis — making alpha-lactalbumin the committed, rate-defining control point of lactose biosynthesis.